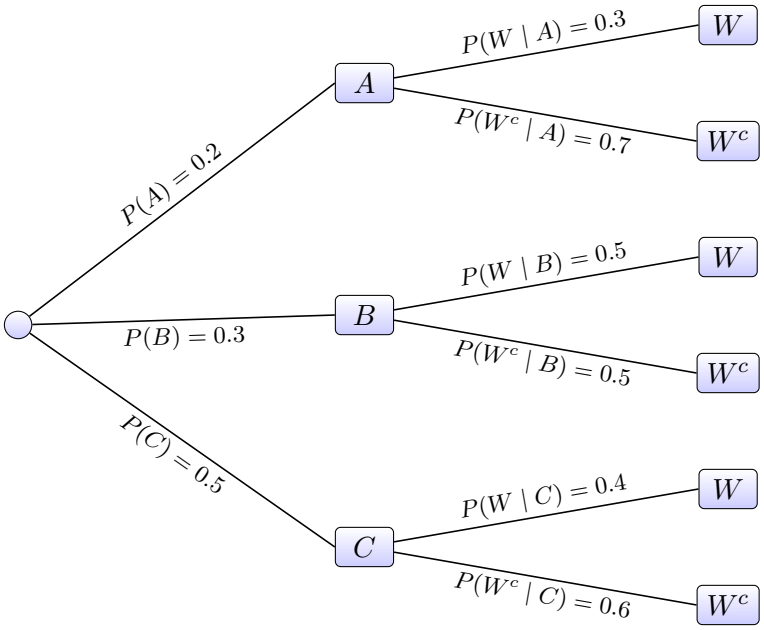
F78PA Probability & Statistics A

Malaysia Class test Solution

1. (a) We can visualise the different outcomes and their probabilities in the following probability tree.



From this probability tree we can calculate the probability that a randomly selected voter from the state will vote for W:

$$\begin{aligned}\mathbb{P}(\text{Vote for } W) &= \mathbb{P}((\text{A}, W)) + \mathbb{P}((\text{B}, W)) + \mathbb{P}((\text{C}, W)) \\ &= \mathbb{P}(A) \cdot \mathbb{P}(W \mid A) + \mathbb{P}(B) \cdot \mathbb{P}(W \mid B) + \mathbb{P}(C) \cdot \mathbb{P}(W \mid C) \\ &= (0.2)(0.3) + (0.3)(0.5) + (0.5)(0.4) \\ &= 0.06 + 0.15 + 0.20 \\ &= 0.41\end{aligned}$$

(2 marks)

(b) By the definition of conditional probability

$$\mathbb{P}(\text{A}|W) = \frac{\mathbb{P}(\text{A and } W)}{\mathbb{P}(W)} = \frac{(0.2)(0.3)}{0.41} = \frac{6}{41}.$$

(2 marks)

(c) Two events E and F are independent if and only if $\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$.

(1 mark)

(for correct definition or correct use of the definition)

Using part (a) and the probability tree we have

$$\begin{aligned}\mathbb{P}(\text{Vote for } W) &= 0.41 \\ \mathbb{P}(\text{A}) &= 0.2 \\ \mathbb{P}(\text{Vote for } W \text{ and from City A}) &= 0.06\end{aligned}$$

Since

$$\mathbb{P}(\text{Vote for } W)\mathbb{P}(A) = (0.41)(0.2) = 0.082 \neq 0.06 = \mathbb{P}(\text{Vote for } W \text{ and from City A}),$$

the events "Vote for W" and "Voter from City A" are not independent.

(2 marks)

(d) We have to find the conditional probability that the voter is from City B given that the voter voted for W and is not from City A. The probability to find is $\mathbb{P}(B \mid W \cap A^c)$.

We have

$$\begin{aligned}
\mathbb{P}(B \mid W \cap A^c) &= \frac{\mathbb{P}(B \cap W \cap A^c)}{\mathbb{P}(W \cap A^c)} \\
&= \frac{\mathbb{P}(B \cap W)}{\mathbb{P}(W \cap A^c)} \quad \Big[\text{since } B \cap W \cap A^c = B \cap W \Big] \\
&= \frac{\mathbb{P}(B \cap W)}{\mathbb{P}(W \cap B) + \mathbb{P}(W \cap C)} \\
&= \frac{(0.3)(0.5)}{(0.3)(0.5) + (0.5)(0.4)} \\
&= \frac{0.15}{0.15 + 0.20} \\
&= \frac{15}{35} = 3/7
\end{aligned}$$

(3 marks)

[Total 10 marks]

2. (a) A function f_T is a valid probability density function if $f_T(t) \geq 0$ for all $t \in \mathbb{R}$ and

$$\int_{-\infty}^{\infty} f_T(t)dt = 1.$$

(1 mark)

To find c, we have

$$1 = \int_{-\infty}^{\infty} f_T(t)dt = \int_0^2 ct^2\, dt = c\left[\ \frac{1}{3}t^3\ \Big|_{t=0}^{t=2}\ \right] = \frac{8c}{3}$$
$$\implies c = 3/8.$$

(1 mark)

(b) The cumulative distribution function F_T is defined by $F_T(t) = \mathbb{P}(T \leq t)$. We observe that since $f_T(t) = 0$ for $t < 0$ we have $F_T(t) = 0$ for $t < 0$ and since $f_T(t) = 0$ for $t > 2$ we have $F_T(t) = 1$ for $t > 2$.

(1 mark)

For $t \in [0, 2]$ we have

$$F_T(t) = \int_{-\infty}^{t} f_T(s)ds = \frac{3}{8}\int_0^t s^2\, ds = \frac{3}{8}\left[\ \frac{1}{3}s^3\ \Big|_{s=0}^{s=t}\ \right] = \frac{t^3}{8}$$

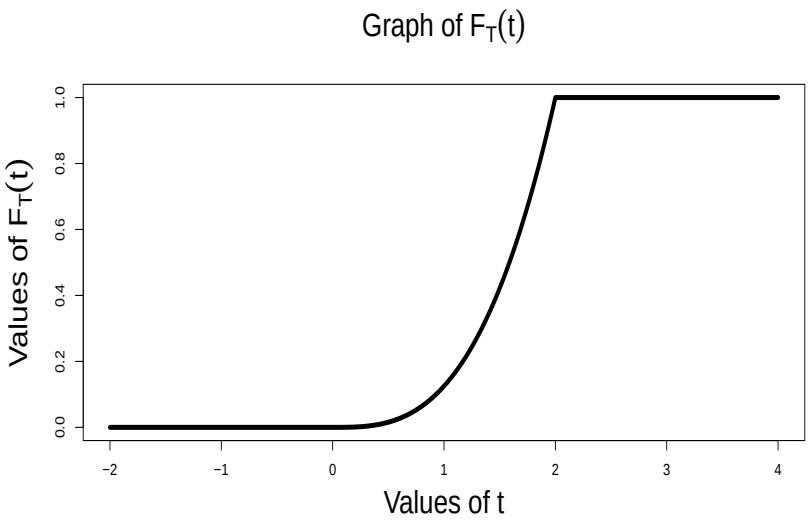
(1 mark)

Therefore

$$F_T(t) = \begin{cases} 0 & \text{for } t < 0, \\ t^3/8 & \text{for } 0 \leq t \leq 2, \text{ and} \\ 1 & \text{for } t > 2. \end{cases}$$

(1 mark)

Sketch of $F_T(t)$:



(1 mark)

(c) The probability that a customer waits between 0.5 and 1.5 minutes is

$$\mathbb{P}(0.5 < T < 1.5) = F_T(1.5) - F_T(0.5) = \frac{1.5^3}{8} - \frac{0.5^3}{8} = 0.40625.$$

(2 marks)

(d) We need to solve the equation below for w:

$$P(T \leq w) = 0.75$$

That is,

$$w^3/8 = 0.75 \implies w^3 = 6 \implies w = 6^{1/3} = 1.82$$

(2 marks)

[Total 10 marks]